

**DRAGOMAN, Bulgaria**WMO: **156050**Lat: **42.933N**Lon: **22.933E**Elev: **716**StdP: **93.01**Time Zone: **2.00 (EUE)**Period: **94-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.1      | -10.8 | -15.8                         | 1.0 | -11.9 | -13.5 | 1.3 | -9.7 | 11.1                  | -1.0 | 9.8 | -2.9 | 2.0                   | 90   | 0.545 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.3                   | 30.9            | 20.0 | 29.1 | 19.5 | 27.4 | 18.7 | 21.1                | 28.2 | 20.2 | 27.4 | 19.3 | 26.2 | 2.8                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.8                            | 14.9 | 24.1 | 17.8 | 13.9 | 23.0 | 16.9 | 13.1 | 22.2 | 64.4          | 28.5 | 61.0 | 27.6 | 58.1 | 26.1 | 25.7              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |  |  |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|--|--|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |  |  |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         |  |  |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |  |  |
| (4) <b>10.1</b>   | <b>8.8</b> | <b>7.4</b> | <b>-19.1</b>               | <b>34.0</b> | <b>3.5</b>         | <b>1.7</b> | <b>-21.6</b>                                       | <b>35.2</b> | <b>-23.7</b> | <b>36.2</b> | <b>-25.7</b> | <b>37.2</b> | <b>-28.2</b> | <b>38.4</b> |  |  |
| (5)               |            |            | <b>-19.3</b>               | <b>23.1</b> | <b>3.4</b>         | <b>1.2</b> | <b>-21.7</b>                                       | <b>23.9</b> | <b>-23.7</b> | <b>24.6</b> | <b>-25.6</b> | <b>25.3</b> | <b>-28.1</b> | <b>26.2</b> |  |  |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.3    | -1.5  | -0.4  | 4.2   | 8.8   | 14.0  | 17.3  | 19.7  | 19.6  | 15.3  | 10.0  | 4.8   | -0.4  |
|                                                                                  | DBStd       | 8.59   | 4.85  | 4.94  | 4.61  | 3.92  | 3.39  | 3.27  | 2.88  | 2.90  | 3.63  | 4.31  | 4.48  | 4.93  |
|                                                                                  | HDD10.0     | 1447   | 355   | 291   | 185   | 64    | 7     | 1     | 0     | 0     | 3     | 55    | 163   | 323   |
|                                                                                  | HDD18.3     | 3444   | 613   | 524   | 439   | 287   | 137   | 57    | 18    | 19    | 104   | 259   | 406   | 581   |
|                                                                                  | CDD10.0     | 1206   | 0     | 0     | 4     | 28    | 131   | 220   | 301   | 298   | 161   | 54    | 8     | 1     |
|                                                                                  | CDD18.3     | 161    | 0     | 0     | 0     | 0     | 3     | 26    | 61    | 58    | 12    | 1     | 0     | 0     |
|                                                                                  | CDH23.3     | 1825   | 0     | 0     | 0     | 8     | 62    | 302   | 639   | 637   | 167   | 11    | 0     | 0     |
|                                                                                  | CDH26.7     | 521    | 0     | 0     | 0     | 1     | 6     | 64    | 203   | 203   | 42    | 2     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.2    | 3.2   | 3.7   | 3.8   | 3.6   | 3.2   | 2.7   | 2.8   | 2.7   | 2.9   | 2.9   | 3.1   | 3.3   |
| Precipitation                                                                    | PrecAvg     | 609    | 39    | 39    | 38    | 54    | 71    | 71    | 54    | 53    | 50    | 54    | 41    | 46    |
|                                                                                  | PrecMax     | 885    | 101   | 71    | 71    | 100   | 161   | 148   | 110   | 172   | 166   | 115   | 122   | 95    |
|                                                                                  | PrecMin     | 369    | 8     | 8     | 6     | 11    | 20    | 20    | 9     | 8     | 8     | 6     | 1     | 9     |
|                                                                                  | PrecStd     | 130    | 25    | 18    | 17    | 24    | 31    | 34    | 28    | 43    | 35    | 34    | 26    | 22    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.2  | 15.1  | 19.7  | 24.2  | 27.4  | 30.6  | 34.0  | 33.4  | 30.3  | 24.6  | 19.6  | 13.5  |
|                                                                                  |             | MCWB   | 7.9   | 9.6   | 12.4  | 15.4  | 17.4  | 20.5  | 19.9  | 20.5  | 19.3  | 16.5  | 13.3  | 10.2  |
|                                                                                  | 2%          | DB     | 9.2   | 11.6  | 17.0  | 19.9  | 25.0  | 28.5  | 31.1  | 31.0  | 27.6  | 21.7  | 16.5  | 10.8  |
|                                                                                  |             | MCWB   | 5.6   | 7.9   | 10.3  | 12.7  | 16.6  | 20.1  | 20.0  | 20.0  | 18.0  | 14.6  | 11.4  | 7.9   |
|                                                                                  | 5%          | DB     | 7.1   | 9.3   | 14.5  | 17.7  | 23.1  | 26.7  | 29.0  | 29.0  | 24.9  | 19.4  | 13.8  | 8.8   |
|                                                                                  |             | MCWB   | 5.0   | 6.4   | 9.0   | 11.2  | 16.1  | 19.2  | 19.5  | 19.2  | 16.6  | 13.5  | 9.9   | 6.9   |
|                                                                                  | 10%         | DB     | 5.1   | 6.9   | 12.1  | 15.8  | 21.4  | 24.8  | 27.0  | 27.0  | 22.5  | 17.2  | 11.7  | 6.8   |
|                                                                                  |             | MCWB   | 3.5   | 4.7   | 7.7   | 10.4  | 15.3  | 18.4  | 18.8  | 18.3  | 15.5  | 12.5  | 8.6   | 5.4   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.2   | 10.7  | 13.3  | 15.8  | 18.7  | 22.1  | 22.4  | 21.9  | 19.6  | 17.3  | 13.8  | 10.4  |
|                                                                                  |             | MCDB   | 11.7  | 13.2  | 18.4  | 23.6  | 25.1  | 27.2  | 29.7  | 29.1  | 28.9  | 23.3  | 18.6  | 13.0  |
|                                                                                  | 2%          | WB     | 6.3   | 8.2   | 11.0  | 13.4  | 17.6  | 20.9  | 20.9  | 20.7  | 18.4  | 15.3  | 11.9  | 8.3   |
|                                                                                  |             | MCDB   | 8.4   | 11.3  | 16.2  | 18.4  | 23.2  | 26.9  | 28.2  | 28.9  | 26.2  | 20.1  | 15.6  | 10.3  |
|                                                                                  | 5%          | WB     | 5.0   | 6.5   | 9.4   | 12.0  | 16.5  | 19.8  | 20.1  | 19.8  | 17.2  | 14.1  | 10.5  | 6.9   |
|                                                                                  |             | MCDB   | 7.0   | 9.1   | 13.8  | 16.5  | 21.9  | 25.5  | 27.3  | 27.7  | 23.7  | 18.1  | 13.2  | 8.6   |
|                                                                                  | 10%         | WB     | 3.6   | 4.9   | 8.0   | 10.8  | 15.5  | 18.8  | 19.2  | 18.8  | 16.1  | 13.0  | 8.9   | 5.4   |
|                                                                                  |             | MCDB   | 5.0   | 6.7   | 11.7  | 15.0  | 20.8  | 24.2  | 25.8  | 25.8  | 21.4  | 16.4  | 11.2  | 6.8   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.9   | 7.8   | 9.6   | 10.1  | 10.8  | 11.3  | 12.3  | 12.5  | 11.5  | 10.2  | 8.8   | 6.4   |
|                                                                                  |             | MCDBR  | 10.3  | 12.7  | 15.0  | 14.5  | 14.3  | 15.1  | 16.0  | 16.7  | 16.2  | 14.1  | 13.2  | 9.2   |
|                                                                                  |             | MCWBR  | 7.5   | 9.5   | 9.6   | 7.9   | 7.7   | 7.7   | 6.9   | 7.2   | 8.0   | 8.1   | 8.6   | 6.8   |
|                                                                                  | 5% WB       | MCDBR  | 9.9   | 12.2  | 13.9  | 12.7  | 13.2  | 14.1  | 14.1  | 14.5  | 14.6  | 12.4  | 11.3  | 8.1   |
|                                                                                  |             | MCWBR  | 7.4   | 9.4   | 9.4   | 7.5   | 7.5   | 7.8   | 6.8   | 6.8   | 7.6   | 7.5   | 7.6   | 6.1   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.312 | 0.341 | 0.384 | 0.442 | 0.430 | 0.422 | 0.427 | 0.440 | 0.413 | 0.372 | 0.338 | 0.313 |
|                                                                                  | taud        |        | 2.346 | 2.281 | 2.220 | 2.069 | 2.168 | 2.229 | 2.216 | 2.179 | 2.232 | 2.339 | 2.399 | 2.386 |
|                                                                                  | Ebn at Noon |        | 815   | 847   | 850   | 824   | 846   | 852   | 843   | 818   | 811   | 802   | 778   | 779   |
|                                                                                  | Edh at Noon |        | 82    | 104   | 125   | 157   | 146   | 138   | 138   | 139   | 122   | 97    | 78    | 73    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1.54  | 2.39  | 3.42  | 4.45  | 5.36  | 6.11  | 6.30  | 5.66  | 4.12  | 2.83  | 1.76  | 1.31  |
|                                                                                  | RadStd      |        | 0.23  | 0.32  | 0.41  | 0.54  | 0.48  | 0.53  | 0.46  | 0.49  | 0.34  | 0.37  | 0.20  | 0.19  |